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Anterior Cruciate Ligament Injuries in National Football League Players from 2012 to 2022: A Descriptive Epidemiology Study

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Abstract:

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Thieme

There is a high incidence of anterior cruciate ligament (ACL) tears in National Football League (NFL) players. There is a relative paucity of long-term data characterizing the effects of artificial versus natural playing surfaces on ACL tear incidence. A comprehensive search of publicly available data identified any NFL player who sustained an ACL injury from 2012 to 2022. The publicly available databases allowed for characterization of playing surface, player position, timing of injury in season, mechanism of injury, and player age. Incidence of ACL injuries by playing surface, player position, and position group relative to all other positions or position groups were compared by chi squared test. Correlates of in-game ACL injury were determined by stepwise multivariate logistic regression. There were 520 NFL players who experienced an ACL tear between seasons 2012-13 and 2022-23. There was no significant difference in incidence of ACL tears on artificial turf versus natural playing surfaces (265 versus 255, p=0.670). Wide receivers (p=0.027) were associated with a significantly higher incidence of ACL tears compared to other positions. Receivers (wide receivers and tight ends) (p=0.007) were associated with significantly higher incidence of ACL tears compared to other position groups, while specialists (punters, kickers, and long snappers) (p<0.001) were associated with significantly lower incidence of ACL tears compared to other position groups. No significant difference between mechanism of injury between playing surfaces was found (p=0.765). More ACL tears occurred before week 9 than on or after week 9 in-season, and the proportion of ACL tears occurring during a game (67%) was significantly higher than those occurring during practice. There was no statistically significant difference in incidence of ACL tears in NFL players from the 2012-13 to 2022-23 seasons based on playing surface (artificial versus natural). Wide receivers had a significantly higher incidence of ACL tears compared to other positions and tears were more common during games and before week 9 in-season.

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Anterior Cruciate Ligament Injuries in National Football League Players from 2012 to 2022: A Descriptive Epidemiology Study

Abstract

There is a high incidence of anterior cruciate ligament (ACL) tears in National Football League (NFL) players. There is a relative paucity of long-term data characterizing the effects of artificial versus natural playing surfaces on ACL tear incidence. A comprehensive search of publicly available data identified any NFL player who sustained an ACL injury from 2012 to 2022. The publicly available databases allowed for characterization of playing surface, player position, timing of injury in season, mechanism of injury, and player age. Incidence of ACL injuries by playing surface, player position, and position group relative to all other positions or position groups were compared by chi squared test. Correlates of in-game ACL injury were determined by stepwise multivariate logistic regression. There were 520 NFL players who experienced an ACL tear between seasons 2012-13 and 2022-23. There was no significant difference in incidence of ACL tears on artificial turf versus natural playing surfaces (265 versus 255, $p=0.670$). Wide receivers ($p=0.027$) were associated with a significantly higher incidence of ACL tears compared to other positions. Receivers (wide receivers and tight ends) ($p=0.007$) were associated with significantly higher incidence of ACL tears compared to other position groups, while specialists (punters, kickers, and long snappers) ($p<0.001$) were associated with significantly lower incidence of ACL tears compared to other position groups. No significant difference between mechanism of injury between playing surfaces was found ($p=0.765$). More ACL tears occurred before week 9 than on or after week 9 in-season, and the proportion of ACL tears occurring during a game (67%) was significantly higher than those occurring during

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practice. There was no statistically significant difference in incidence of ACL tears in NFL players from the 2012-13 to 2022-23 seasons based on playing surface (artificial versus natural). Wide receivers had a significantly higher incidence of ACL tears compared to other positions and tears were more common during games and before week 9 in-season.

Keywords: anterior cruciate ligament; National Football League; natural grass; artificial turf; knee injury

Introduction

Anterior cruciate ligament (ACL) injuries are among the most prevalent sports-related injuries; an estimated 300,000 are diagnosed annually in the US.[1] Knee injuries make up 29.3% of lower extremity injuries in the National Football League (NFL), with ACL injuries making up 2% of all NFL injuries.[2,3] Return to sport after treatment of an ACL tear typically requires more than 9 months, and 20% of NFL running backs and wide receivers who sustain an ACL injury never return to play in the NFL. For the 80% that do return to play, their on-field production is decreased by 33%.[4] It is also reported that 35.7% of offensive and defensive lineman who tear their ACL never returned to play in the NFL.[5] Secrist et al. reported that NFL players who do return to sport earned, on average, $2,070,521 less than their non-injured counterparts four years following an ACL tear.[6] As such, these injuries can be physically, mentally, and economically devastating to NFL players and warrant further investigation to characterize risk factors and determine optimal prevention strategies.

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Risk factors for injuries to NFL players are a growing focus, with equipment modifications and implemented rule changes having the potential to limit injury exposure and increase player safety. In addition, recent attention has centered on the potential effects of artificial surfaces compared to natural grass on the development of lower extremity injuries, specifically, ACL tears. Both the National Collegiate Athletic Association (NCAA) and the NFL data provide some evidence for increased ACL injuries on artificial surfaces compared to natural grass.[7,8] Discussions regarding the possible links between artificial playing surfaces and non-contact ACL injuries in NFL players have intensified, leading to the NFL Players Association requesting that NFL teams switch to natural grass playing surfaces.[9–12] However, current evidence is mixed, with some studies reporting higher incidence of ACL injuries in NCAA college football players and NFL players competing on artificial playing surfaces while other studies have reported no increased risk of ACL injury associated with artificial playing surfaces.[7,13–16] Importantly, the aforementioned studies have not determined incidence of ACL tears on artificial versus natural grass playing surfaces over 10 or more NFL seasons nor fully characterized ACL injuries by player position, timing, or mechanism of injury.[8,14] An analysis of ACL tears from 2010 to 2013 showed that wide receivers, running backs, tight ends, full backs, and linebackers face significantly higher risk for ACL tears compared to other positions.[14] Another study from 2015 to 2019 found that preseason games are associated with increased ACL injury rates compared to regular season and postseason NFL games.[8]

Based on the need for a more comprehensive analysis, this study was designed to determine the incidence of ACL injuries in NFL players between the 2012-13 to 2022-23 seasons on artificial versus natural grass playing surfaces while also characterizing ACL tears occurring during this

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time period based on player position, timing in season, and contact versus non-contact injury mechanism. We hypothesized that a greater number of ACL injuries would occur on artificial playing surfaces compared to natural grass, more injuries would occur earlier in the season, and positions requiring more speed and cutting maneuvers (receiver, running back, linebacker) would have a higher incidence of ACL injury.

Methods

Data Collection

An online search was undertaken to compile a list of ACL tears in NFL players from the 2012-13 to 2022-23 seasons using publicly available data. Data collection started with the 2012-13 season due to a league-wide lockout prior to the 2011-12 season, affecting injury data. Publicly available resources including Pro-Football-Reference.com, ESPN.com, NFL game films from 2012-13 to 2022-23, and NFL team injury reports were used to compile the dataset.[17–19] For each individual season, searches for "knee injury," "ACL," and "anterior cruciate ligament" were conducted to procure the injury data from the online websites and team injury reports. Player demographic data, team, position, age at time of injury, and seasons played was collected from Pro-Football-Reference.com.[19] NFL injury reports, NFL.com, and other publicly available sources were used to identify which week within the season an ACL tear occurred. Mechanism of injury was determined by watching NFL game film. News reports were reviewed if the injury occurred outside of camera view during NFL games and could not be visualized. News reports were also used for injuries occurring during practice since practice film is not available for public use. News reports for an injury were used if they explicitly stated if the injury occurred due to a contact or non-contact injury mechanism. A contact injury was defined as any contact

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occurring immediately prior to or at the time of ACL tear. A non-contact injury was defined as an injury with no contact immediately prior to the ACL tear. ACL tears were documented to occur at practice or a game, and timing in season was recorded. Only football related injuries were included, and all ACL tears outside of football related activities were excluded. Playing surface type was determined as grass or artificial turf from NFL.com and official NFL team websites at the time of injury. The playing surface was identified and recorded for each NFL game from 2012-13 to 2022-23. Player positions were grouped based on previously published literature.[14] Position groups consisted of receivers (wide receivers and tight ends), backs (linebackers, running backs, and fullbacks), defensive backs (cornerbacks and safeties), perimeter linemen (defensive ends and offensive tackles), quarterbacks, interior linemen (guards, centers, and defensive tackles), and specialists (punters, kickers, and long snappers). Incidence of ACL injuries per game was determined such that a game played would count as 1 game played for both teams for each rostered player.

Power Analysis

Power calculation was conducted to determine a 2% difference incidence rate among 2 independent samples with an incidence rate in group 1 of 4%, incidence rate in group 2 of 6%, alpha of 0.05, beta of 0.2 and power of 0.8, yielding a sample size of 3,726. In the current study 5,948 total games split approximately evenly between the two playing surfaces exceeds the sample size requirements for detecting a 2% difference in incidence rate.

Statistical Analysis

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The incidence of ACL injuries by playing surface, player position and position group relative to all other positions or position groups were compared by chi-squared test. Correlates of in-game ACL injury were determined by stepwise multivariate logistic regression of each rostered player appearing in at least 1 game, with initial independent variables of playing surface, season timing, position, body mass index (BMI), age, total games played in that season for the individual player, and player experience. The regression was repeated with stepwise elimination of the least correlated independent variable by corrected p value until only significant correlatives remained following Holm correction for multiple hypothesis testing. Subsequent testing for multicollinearity was undertaken to reduce variables with substantial correlation amongst themselves. Overfitting was reduced by ensuring there would be >15 outcomes of interest for each independent variable included in the regression model. Data manipulation and statistical analysis were performed using R software.[20] A p value of <0.05 was set for significance.

Results

*ACL Tears by Season (**Table 1**)*

From NFL seasons 2012-13 to 2022-23, there were 520 total ACL tears. Overall, there was no significant difference in injuries sustained playing on artificial turf versus natural grass (265 versus 255, respectively, p=0.670). (%) indicates the proportion of ACL tears each season on grass, artificial turf, or both to the total number of ACL tears on the respective playing surface.

*ACL Tears by Week (**Table 2**)*

There were increased number of ACL tears at the beginning of the season (<9 weeks) relative to later in the season. A total of 291 ACL tears prior to week 9 and 126 ACL tears during or after

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week 9. The early season period experienced the highest number of ACL tears. A total of 132 ACL tears occurred during the pre-season.

*Biodemographic Data for Players Tearing their ACL (**Table 3**)*

There were no differences in biodemographic variables among players sustaining ACL tears on grass compared to artificial turf. The average BMI of a player with a torn ACL was 30.6 (±4.2) kg/m^2, at an average age of 25.2 yrs. Players had an average of 2.7 years of NFL experience at the time of ACL injury.

*ACL Tears by Mechanism on Playing Surface (**Table 4**)*

Of the total 520 ACL tears, 245 tears were due to a contact injury mechanism. A total of 172 were due to a non-contact mechanism with 113 tears, 46 game tears, and 67 practice tears, due to unknown mechanism. A total of 57.8% of all ACL tears occurred on natural grass due to a contact mechanism with 42.2% having occurred due to a non-contact mechanism. Similarly, 59.6% of all ACL tears on artificial turf surfaces occurred due to a contact mechanism, while 40.4% occurred due to a non-contact mechanism. There was no difference between mechanism of injury between playing surfaces (p=0.702).

*ACL Tears by Practice or Game (**Table 5**)*

ACL tears sustained during game play had over double the amount of ACL tears (352, 67.7%) versus injuries sustained during practice (168, 32.3%), despite having far fewer games than practices over the course of a season. On both natural grass and artificial turf, players tore their ACL 84 times while practicing. Players tore their ACL on grass and turf 171 and 181 times,

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respectively, during games. Differences in the number of ACL tears that occurred during practice or during games was not found to be significant (p=0.762).

*Incidence of ACL Injuries by Playing Surface (**Table 6**)*

There was no significant difference in incidence of ACL injury by playing surface as calculated by game incidence rates (0.058 per game on grass versus 0.061 per game on artificial turf, p = 0.661). While incidence rates for practice could not be calculated due to unknown total number of practices, 168 ACL injuries occurred during practice, split evenly between grass and artificial turf playing surfaces.

*Incidence of ACL Injuries by Player Position (**Table 7**)*

Defensive backs sustained the highest total amount of ACL tears (n=105), but not significantly more when stratified by number of players per position per season (0.022 overall, p=0.781). Fullbacks (0.039 overall, p = 0.215) and wide receivers (0.028 overall, p = 0.027) sustained the highest incidence of ACL injuries when compared relative to all other positions. Kickers had the lowest total number of ACL injuries (one) and a significantly lower relative incidence of per player-season ACL tears versus other positions (0.002 overall, p < 0.001).

*Incidence of NFL Player ACL Injuries by Position Group (**Table 8**)*

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Receivers (0.028, $p = 0.007$), which includes wide receivers and tight ends, had the highest incidence of ACL injuries when compared relative to all other position groups. Specialists (0.007, $p < 0.001$), which includes punters, kickers and long snappers, had the lowest overall incidence of ACL injuries.

Discussion

For the 2012-13 to 2022-23 NFL seasons, there was no significance difference in incidence of ACL tears sustained during games on artificial (0.058) versus natural grass (0.061) surfaces. There were, however, significant differences in ACL injury rates based on position.

Wide receivers were associated with a significantly higher incidence of ACL tears compared to other positions (0.028 vs. 0.022), and receivers (wide receivers and tight ends) were associated with significantly higher incidence of ACL tears compared to other position groups (0.028 vs. 0.022). Specialists (punters, kickers, and long snappers) were associated with significantly lower incidence of ACL tears compared to other position groups (0.007 vs. 0.022).

The findings—which demonstrate no significant difference in ACL injury incidence based on playing surface type over the 2012-13 to 2022-23 NFL seasons—are consistent with previous older, shorter term studies.[2,13,14] While multiple studies also reported no significant differences in ACL tear incidence based on playing surface type for NCAA football players, Loughran et al. reported that Division II and III college football players sustained ACL injuries at significantly higher rates on artificial versus natural playing surfaces.[7,16,21] Hershman et al. reported a significant increase in ACL injury rate on the artificial surface FieldTurf compared to natural

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grass.[15] These disparate reports can be attributed to two main factors: number of seasons analyzed, older playing surfaces, or a combination of both. Loughran et al. specifically analyzed college football players from the 2004 to 2014 seasons and Hershman et al. analyzed NFL players from the 2000 to 2009 seasons. Since the early 2000s, the quality, structure, and composition of artificial surfaces has improved.[22] Studies in NCAA and NFL players that reported similar findings analyzed between 3 to 5 seasons of injury data. Our study window of 11 seasons provides a longer-term accounting of ACL injuries in NFL players.

The findings from the present study regarding higher incidence of ACL tears early in the NFL season with a progressive decline as the season progresses parallel the data reported by Dodson et al.[14] This pattern has been attributed to larger NFL roster sizes during training camp and the preseason, less strenuous and less frequent practices as the season progresses, and improved player conditioning levels later in-season. During the 2020-21 NFL season—which did not include a preseason period due to the COVID-19 pandemic—the frequency of ACL tears was highest in September. In seasons prior to and after 2020-21 for the present study, the highest frequency of ACL tears occurred in August. The proportions of ACL tears during the first month of play was not different during the study time period when the first month was September or August.[23] In 2021, the NFL removed the fourth preseason game and added a week 18 game to the season at least in part as a response to these findings.[24]

During the study period, more ACL tears were reported to occur in association with a contact mechanism of injury rather than a non-contact mechanism of injury, but these differences did not reach statistical significance. The injury mechanism differences held true for artificial and

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natural playing surfaces. These findings are in contrast to some previous reports, including a video analysis study of ACL tears in NFL players from 2013 to 2016 that documented over 70% of ACL tears resulting from a noncontact mechanism.[25] These contrasting results may be attributed to smaller sample size (3 seasons vs. 11 seasons) and the exclusion of ACL tears that were not able to be video analyzed in the previous study.

Our analysis of practice and game related ACL tears showed that 68% of ACL tears occurred in games while 32% of tears occurred during practice. This is similar to an epidemiology study characterizing ACL tears during the 2013-14 to 2019-20 NFL seasons, which showed 67.6% of ACL tears occur during games versus 31.1% of ACL tears occurring during practice.[23]

Between the 2012-13 to 2022-23 NFL seasons, wide receivers experienced significantly higher rates of ACL tears compared to other positions when analyzed per player-season (78 ACL tears over 2,780 player-seasons). Wide receivers often perform rapid change of direction movements, are involved in high-speed tackles/collisions, and often play on special teams. These factors likely increase their risk for ACL tears.[2,8,26,27] NFL teams differ in their use of the fullback position. Some teams designate the fullback as a primary blocking back for the running back, while other teams use the fullback as a running back/tight end hybrid player. Several NFL teams do not incorporate a fullback in the majority of their play formations. Fullbacks from the 2012-13 to 2021-22 seasons suffered 5 total ACL tears while playing a total of 130 player-seasons. The next lowest number of player-seasons for an individual position were long snappers with 335. Long snappers only enter the game for special teams scenarios including field-goal attempts, point-after-touchdown plays, and punts. Their use is very limited and there have been

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different rules in place to protect them from injury. Some teams use centers as long snapper rather than have a dedicated long snapper. Due to the limited use of long snappers, this position has fewer exposures each game compared to other non-special team positions, explaining the low number of ACL tears and player seasons. NFL kickers experienced a significantly lower total number of ACL tears compared to the rest of the NFL players, and a significantly lower incidence relative to per player-season (1 ACL tear over 502 player-seasons). The significantly lower incidence can be attributed to fewer kickers that play a fewer number of plays throughout the course of a game, only entering the game for kickoffs, field-goal attempts, and point-after-touchdown plays. Penalties such as 'running into the kicker' and 'roughing the kicker' deter players from making physical contact with kickers. The combination of low game exposure and rules protecting other players from making contact with kickers may contribute to why kickers have a significantly lower incidence rate of ACL tears compared to the rest of the NFL.

When player positions were grouped together, receivers (wide receivers and tight ends) sustained ACL tears at higher rates compared to the rest of the NFL players. Specialists (punters, kickers, and long snappers) sustained ACL tears at lower overall rates compared to the rest of the NFL players. Interestingly, Dodson et al. reported that receivers (WR and TE) and backs (LB, RB, FB) suffered ACL tears at higher rates than other NFL position groups and that perimeter linemen (DE and OT) sustained ACL tears at significantly lower rates than other NFL positions between the 2010-11 to 2012-13 seasons.[14] The discrepancy in findings between the aforementioned study and the current study may be attributed to the difference in study window length. The aforementioned study characterized ACL tears over 3 seasons compared to 11 seasons. The longer study window in the current study increases the generalizability of the study.

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It is also important to note that the NFL lockout occurred prior to the 2011-12 season, which prohibited organized team events and did not allow for a normal off-season training period. This lockout potentially had an impact on perimeter linemen.

Limitations

The data from this study were collected from publicly accessible databases and NFL injury reports and therefore may not be a complete representation of all ACL injuries sustained during the study time period.[28,29] However, an ACL injury is typically season ending, and as such is typically reported when it occurs. Given this information, it is unlikely there would be a significant amount of unreported ACL injuries in the NFL. Another limitation of the study is there was not a reported or searchable mechanism for every injury reported. Not all ACL tears were able to be assessed for mechanism due to a lack of film showing tears at practice or if an ACL tear occurred outside of camera view in a game setting. News reports the mechanism of injury for ACL tears during practice or it reports when an ACL tear occurred outside of camera view during a game; these might not be accurate in all cases and pose as a limitation for the current study. Playing surface was sometimes not able to be assessed if teams practiced on both natural grass and artificial turf field during the same day. Finally, sometimes players change positions or also play on special teams, which could affect the injury rate per player position data. To mitigate this limitation the study data was formulated based on the players' listed position solely at the time of injury.

Conclusion

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There was no statistically significant difference in incidence of ACL tears in NFL players from the 2012-13 to 2022-23 seasons based on playing surface (artificial versus natural). Wide receivers had a significantly higher incidence of ACL tears compared to other positions and tears were more common during games and before week 9 in-season.

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Table 1. ACL Tears by Season[a]

	Grass (N=255)	**Artificial Turf (N=265)**	**Total (N=520)**	**value**
Season, n (%)				0.670
2012	16 (6.3%)	28 (10.6%)	44 (8.5%)	

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2013	31 (12.2%)	32 (12.1%)	63 (12.1%)
2014	23 (9.0%)	20 (7.5%)	43 (8.3%)
2015	22 (8.6%)	26 (9.8%)	48 (9.2%)
2016	26 (10.2%)	20 (7.5%)	46 (8.8%)
2017	30 (11.8%)	20 (7.5%)	50 (9.6%)
2018	23 (9.0%)	29 (10.9%)	52 (10.0%)
2019	15 (5.9%)	19 (7.2%)	34 (6.5%)
2020	17 (6.7%)	17 (6.4%)	34 (6.5%)
2021	27 (10.6%)	28 (10.6%)	55 (10.6%)
2022	25 (9.8%)	26 (9.8%)	51 (9.8%)

[a]ACL, Anterior Cruciate Ligament

Table 2. ACL Tears by Week[a]

	Grass (N=255)	Artificial Turf (N=265)	Total (N=520)	
Time in Season, n (%)				**value P=0.771**
Pre-season	**67 (33.0%)**	**65 (30.4%)**	**132 (31.7%)**	
Early Season (Week 1-8)	**76 (37.4%)**	**83 (38.8%)**	**159 (38.1%)**	
Late Season (Week 9-18)	**57 (28.1%)**	**60 (28.0%)**	**117 (28.1%)**	
Post-season	**3 (1.5%)**	**6 (2.8%)**	**9 (2.2%)**	

[a]ACL, Anterior Cruciate Ligament

Table 3. Biodemographic Data for Players Tearing their ACL[a]

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	Grass (N=255)	Artificial Turf (N=265)	Total (N=520)	p value
Body Mass Index, n (%)				0.708
18.5 – 24.9	14 (5.5%)	13 (4.9%)	27 (5.2%)	
25.0 – 29.9	122 (47.8%)	114 (43.0%)	236 (45.4%)	
30.0 – 34.9	67 (26.3%)	84 (31.7%)	151 (29.0%)	
35 – 39.9	48 (18.8%)	49 (18.5%)	97 (18.7%)	
40+	4 (1.6%)	5 (1.9%)	9 (1.7%)	
Years Experience, n (%)				0.781
Less than 4 seasons	178 (69.8%)	182 (68.7%)	360 (69.2%)	
4 seasons or more	77 (30.2%)	83 (31.3%)	160 (30.8%)	
Race, n (%)				0.692
Black	183 (71.8%)	190 (71.7%)	373 (71.7%)	
White	66 (25.9%)	70 (26.4%)	136 (26.2%)	
Hispanic	0 (0.0%)	1 (0.4%)	1 (0.2%)	
Other	6 (2.4%)	4 (1.5%)	10 (1.9%)	

[a]ACL, Anterior Cruciate Ligament

Table 4. ACL Tears by Mechanism on Playing Surface[a]

	Grass (N=199)	Artificial Turf (N=218)	Total (N=417)	value
Mechanism of ACL Tear, n (%)[a]				0.702
Contact	115 (57.8%)	130 (59.6%)	245 (58.8%)	

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Non-contact	84 (42.2%)	88 (40.4%)	172 (41.2%)

[a]ACL, Anterior Cruciate Ligament

Table 5. ACL Tears by Practice or Game[a]

	Grass (N=255)	Artificial Turf (N=265)	Total (N=520)	value
Injury Sustained During Game or Practice, n (%)				0.762
Practice	84 (32.9%)	84 (31.7%)	168 (32.3%)	
Game	171 (67.1%)	181 (68.3%)	352 (67.7%)	

[a]ACL, Anterior Cruciate Ligament

Table 6. Incidence of ACL Injuries by Playing Surface, 2012-2022[a]

Playing Surface	Practice Injuries, n	Game Injuries, n	Total Games, n	Incidence, $game^{-1}$	P Value
Grass	84	171	2968	0.058	0.661
Artificial Turf	84	181	2980	0.061	

[a]ACL, Anterior Cruciate Ligament

Table 7. Incidence of ACL Injuries by Player Position, 2012-2022[a]

Position	Injuries, n	Total Player-seasons, n	Incidence, player-	P Value (vs Rest

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			season^{-1}	of NFL)[b]
Fullback	5	130	0.039	0.215
Wide receiver	78	2780	0.028	**0.027**
Tight end	40	1481	0.027	0.112
Linebacker	80	3301	0.024	0.338
Offensive tackle	38	1611	0.024	0.660
Defensive back	105	4660	0.023	0.781
Quarterback	18	855	0.021	1.000
Running back	40	1903	0.021	0.747
Defensive end	39	1899	0.021	0.628
Center	13	688	0.019	0.692
Guard	25	1338	0.019	0.443
Defensive tackle	31	1731	0.018	0.268
Long Snapper	4	335	0.012	0.260
Punter	3	386	0.008	0.053
Kicker	1	502	0.002	**< 0.001**
All NFL players	520	23600	0.022	

[a]ACL, Anterior Cruciate Ligament, [b]NFL, National Football League, Bold indicates significance

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Table 8. Incidence of NFL Player ACL Injuries by Position Group, 2012-2022[a]

Position Group[b]	Injuries, n	Total Player-seasons, n	Incidence, player-season^{-1}	P Value (vs Rest of NFL)[c]
Receivers (WR, TE)	118	4261	0.028	**0.007**
Backs (LB, RB, FB)	125	5334	0.023	0.427
Defensive backs (CB, S)	105	4660	0.023	0.781
Perimeter linemen (DE, OT)	77	3510	0.022	1.000
Quarterbacks	18	855	0.021	1.000
Interior linemen (G, DT, C)	69	3757	0.018	0.102
Specialists (P, K, LS)	8	1223	0.007	**< 0.001**
All NFL players	520	23600	0.022	

[a]ACL, Anterior Cruciate Ligament, [b]WR, Wide Receiver, TE, Tight End, LB, Linebacker, RB, Running Back, FB, Fullback, CB, Cornerback, S, Safety, DE, Defensive End, OT, Offensive Tackle, G, Guard, DT, Defensive Tackle, C, Center, P, Punter, K, Kicker, LS, Long Snapper, [c]NFL, National Football League

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